

MATH 2211, SPRING 2026: SOLUTIONS TO HOMEWORK 3

- (1) Decide if the following statements are true or false. Give a line or two in justification for your answer. It is permissible for that justification to be a reference to previous homeworks or to lectures:

(a) $\mathbb{Z}/4\mathbb{Z}$ is a field under modular arithmetic.

Solution. False. Not all non-zero elements have multiplicative inverses: for instance, 2 does not. See the table in Lecture 4.

(b) If F is a field and $a, b \in F$ are such that $a \cdot b = 0$, then one of a or b must be 0.

Solution. True. You proved this in Problem 3(a) of Homework 2. See also the general argument for vector spaces in Lecture 7.

(c) The set $\text{Poly}_3(F)$ of polynomials of degree 3 with coefficients of F is a subspace of $\text{Poly}(F)$.

Solution. False. This does not contain the 0 polynomial.

(d) The equation $z^n = a$ for $a \in \mathbb{C}$, $a \neq 0$, admits n distinct solutions in $\mathbb{C}$.

Solution. True. If $z = re^{i\theta}$, then the solutions are $r^{1/n}e^{\frac{i(\theta+2\pi k)}{n}}$ for $0 \leq k \leq n-1$. See Lecture 6.

(e) There exists a complex number z such that $z\bar{z} - 1 = i$.

Solution. False. $z\bar{z} = |z|^2$ is a positive real number for any complex number z ; see p. 10 of Lecture 5.

- (2) Decide if the following subsets are $\mathbb{R}$ -vector subspaces. Again, justify your responses.¹

(a) $U_1 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 + 2x_2 + 3x_3 = 0\} \subset \mathbb{R}^3$;

Solution. This is a subspace. We have to check:

- (Contains 0): The zero vector is $(0, 0, 0)$, and this clearly satisfies the given equation.
- (Closure under +): If $(x_1, x_2, x_3), (y_1, y_2, y_3)$ are in U_1 , then we have

$$(x_1 + y_1) + 2(x_2 + y_2) + 3(x_3 + y_3) = (x_1 + 2x_2 + 3x_3) + (y_1 + 2y_2 + 3y_3) = 0 + 0 = 0.$$

This shows that $(x_1 + y_1, x_2 + y_2, x_3 + y_3) \in U_1$.

- (Closure under scaling): If $(x_1, x_2, x_3) \in U_1$ and $c \in F$, then

$$cx_1 + 2(cx_2) + 3(cx_3) = cx_1 + c(2x_2) + c(3x_3) = c(x_1 + 2x_2 + 3x_3) = c \cdot 0 = 0.$$

Note that we are using several properties of fields and vector spaces here: To know that $2(cx_2) = c(2x_2)$ for instance, we are using the equalities:

$$2(cx_2) = (2c)x_2 = (c \cdot 2) \cdot x_2 = c \cdot (2 \cdot x_2).$$

The first and the last equalities are from the associativity of scalar multiplication in V , and the second equality is commutativity of multiplication in F . Similarly, the property $c \cdot 0 = 0$ was something we showed in Lecture 7.

(b) $U_2 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1x_2x_3 = 0\} \subset \mathbb{R}^3$;

Solution. This is not a subspace. Let's check:

- (Contains 0): The zero vector is $(0, 0, 0)$, and this clearly satisfies the given equation.
- (Closure under +): This fails. For instance $(1, 1, 0) \in U_2$ and $(0, 0, 1) \in U_2$, but their sum $(1, 1, 1)$ is not in U_2 .

(c) $U_3 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 + 2x_2 + 3x_3 = 1\} \subset \mathbb{R}^3$;

¹We will use 'tuple' notation instead of column vector notation to save space.

Solution. This is also not a subspace. Here, it is easy to check that $0 \notin U_3$.

(d) $U_4 = \{f(x) \in \text{Poly}(\mathbb{R}) : f'(-1) = 3f(2)\} \subset \text{Poly}(\mathbb{R});$

Solution. This is a subspace. Let's check:

- (Contains 0): the zero polynomial and its derivative (which is also the zero polynomial) both have value 0 at every point, so the equality $f'(-1) = f(2)$ clearly holds for it.
- (Closure under +): If $f_1(x), f_2(x) \in U_4$, then

$$(f_1 + f_2)'(-1) = f_1'(-1) + f_2'(-1) = 3f_1(2) + 3f_2(2) = 3(f_1 + f_2)(2).$$

What's being used here is that the derivative of a sum of functions is the sum of the derivatives.

- (Closure under scalar multiplication): If $c \in \mathbb{R}$, $f(x) \in U_4$, then

$$(cf)'(-1) = cf'(-1) = c(3f(2)) = 3cf(2).$$

What's being used here is that scalars can be pulled out of the derivative.

(e) $U_5 = \{A \in M_{2 \times 2}(\mathbb{R}) : \det(A) = 0\} \subset M_{2 \times 2}(\mathbb{R});$

Here, $M_{2 \times 2}(\mathbb{R})$ is the $\mathbb{R}$ -vector space of 2×2 -matrices $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\det(A)$ is the *determinant* $ad - bc$.

Solution. This is not a subspace, because it's not closed under addition. if $A_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $A_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$, then $\det(A_1) = \det(A_2) = 0$ while $\det(A_1 + A_2) = 1$.

(3) Let V be an F -vector space.

(a) If $U_1, U_2 \subset V$ are subspaces, show that the subsets

$$U_1 + U_2 = \{u_1 + u_2 : u_1 \in U_1, u_2 \in U_2\} \subset V ; U_1 \cap U_2 \subset V$$

are also subspaces.

Solution. Let us check for $U_1 + U_2$:

- (i) (Contains 0): Since $0 \in U_1$ and $0 \in U_2$, $0 + 0 = 0$ is an element of $U_1 + U_2$.
- (ii) (Closed under +): If $u_1 + u_2 \in U_1 + U_2$ and $v_1 + v_2 \in U_1 + U_2$, then we find

$$\begin{aligned} (u_1 + u_2) + (v_1 + v_2) &= u_1 + (u_2 + (v_1 + v_2)) \\ &= u_1 + ((u_2 + v_1) + v_2) \\ &= u_1 + ((v_1 + u_2) + v_2) \\ &= u_1 + (v_1 + (u_2 + v_2)) \\ &= (u_1 + v_1) + (u_2 + v_2), \end{aligned}$$

where $u_1 + v_1 \in U_1$ and $u_2 + v_2 \in U_2$.

I have carefully used all the axiomatic conditions for a vector space here to show how it's done, but going forward, we can just blindly take on faith that all addition operations in a vector spaces can be done in any order we like: This is a matter of repeatedly combining associativity and commutativity for addition: All the equalities above except for the 3rd use associativity while the 3rd uses commutativity.

- (iii) (Closed under scalar multiplication): If $u_1 + u_2 \in U_1 + U_2$ and $c \in F$, then

$$c \cdot (u_1 + u_2) = c \cdot u_1 + c \cdot u_2$$

where $c \cdot u_1 \in U_1$ and $c \cdot u_2 \in U_2$.

Now, let's do $U_1 \cap U_2$:

- (i) (Contains 0): Since $0 \in U_1$ and $0 \in U_2$, it is contained in both and hence in the intersection.
- (ii) (Closed under +): If $u \in U_1 \cap U_2$ and $v \in U_1 \cap U_2$, then $u + v$ is in both U_1 (because U_1 is a subspace) and also in U_2 . Hence it is contained in $U_1 \cap U_2$.

(iii) (Closed under scalar multiplication): If $u \in U_1 \cap U_2$ and $c \in F$, then $c \cdot u$ is once again in both U_1 and U_2 and hence in $U_1 \cap U_2$.

Note that we have to use the fact that U_1 and U_2 are both subspaces above.

(b) Prove or give a counterexample: If $U_1, U_2, W \subset V$ are all subspaces, and if $U_1 + W = U_2 + W$, then $U_1 = U_2$.

Solution. This is *false* in general. We can take $W = V$, and in this case, $U_1 + W$ and $U_2 + W$ are both just V itself, regardless of what U_1 and U_2 are.

(4) Is $w = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \in \mathbb{C}^3$ a linear combination of the following vectors?

$$\begin{bmatrix} 1 \\ 1 \\ -i \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix}.$$

Solution. We are being asked to solve for $a_1, a_2, a_3 \in \mathbb{C}$ for the equation

$$a_1 \begin{bmatrix} 1 \\ 1 \\ -i \end{bmatrix} + a_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + a_3 \begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

This works out to:

$$\begin{bmatrix} a_1 + 2a_2 + a_3 \\ a_1 + a_2 \\ -ia_1 + ia_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

You should find that this system is *inconsistent*: In matrix notation, what we have is

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 1 & 1 & 0 & -1 \\ -i & 0 & i & 0 \end{array} \right]$$

Using row operations we get

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & -1 & -1 & -2 \\ 0 & 2i & 2i & 1 \end{array} \right] &\rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & i & i & 1/2 \end{array} \right] \\ &\rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1/2 - 2i \end{array} \right] \end{aligned}$$

The last row tells us that the system is inconsistent (why?).

(5) Is the polynomial $x + 1 \in \text{Poly}(\mathbb{R})$ a linear combination of the following polynomials?

$$x^2 + 1, x^3 + x, 2x^2 + x, x + 3 \in \text{Poly}(\mathbb{R}).$$

Solution. We are solving for

$$a_1(x^2 + 1) + a_2(x^3 + x) + a_3(2x^2 + x) + a_4(x + 3) = x + 1,$$

which works out to:

$$a_2x^3 + (a_1 + 2a_3)x^2 + (a_2 + a_3 + a_4)x + (a_1 + 3a_4) = x + 1.$$

Comparing coefficients, we get

$$\begin{bmatrix} a_2 \\ a_1 + 2a_3 \\ a_2 + a_3 + a_4 \\ a_1 + 3a_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

This is the system

$$\begin{bmatrix} 0 & 1 & 0 & 0 & | & 0 \\ 1 & 0 & 2 & 0 & | & 0 \\ 0 & 1 & 1 & 1 & | & 1 \\ 1 & 0 & 0 & 3 & | & 1 \end{bmatrix}$$

Use row operations:

$$\begin{aligned} \begin{bmatrix} 1 & 0 & 2 & 0 & | & 0 \\ 0 & 1 & 0 & 0 & | & 0 \\ 0 & 1 & 1 & 1 & | & 1 \\ 1 & 0 & 0 & 3 & | & 1 \end{bmatrix} &\rightarrow \begin{bmatrix} 1 & 0 & 2 & 0 & | & 0 \\ 0 & 1 & 0 & 0 & | & 0 \\ 0 & 0 & 1 & 1 & | & 1 \\ 0 & 0 & -2 & 3 & | & 1 \end{bmatrix} \\ &\rightarrow \begin{bmatrix} 1 & 0 & 0 & -2 & | & -2 \\ 0 & 1 & 0 & 0 & | & 0 \\ 0 & 0 & 1 & 1 & | & 1 \\ 0 & 0 & 0 & 5 & | & 3 \end{bmatrix} \\ &\rightarrow \begin{bmatrix} 1 & 0 & 0 & -2 & | & -2 \\ 0 & 1 & 0 & 0 & | & 0 \\ 0 & 0 & 1 & 1 & | & 1 \\ 0 & 0 & 0 & 1 & | & 3/5 \end{bmatrix} \\ &\rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & | & -4/5 \\ 0 & 1 & 0 & 0 & | & 0 \\ 0 & 0 & 1 & 0 & | & 2/5 \\ 0 & 0 & 0 & 1 & | & 3/5 \end{bmatrix} \end{aligned}$$

This gives us $a_1 = -4/5$, $a_2 = 0$, $a_3 = 2/5$ and $a_4 = 3/5$.